Ex.No.-1.1-Design and Draft the given 2D Sketches in modelling software.

AIM

To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 2 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE

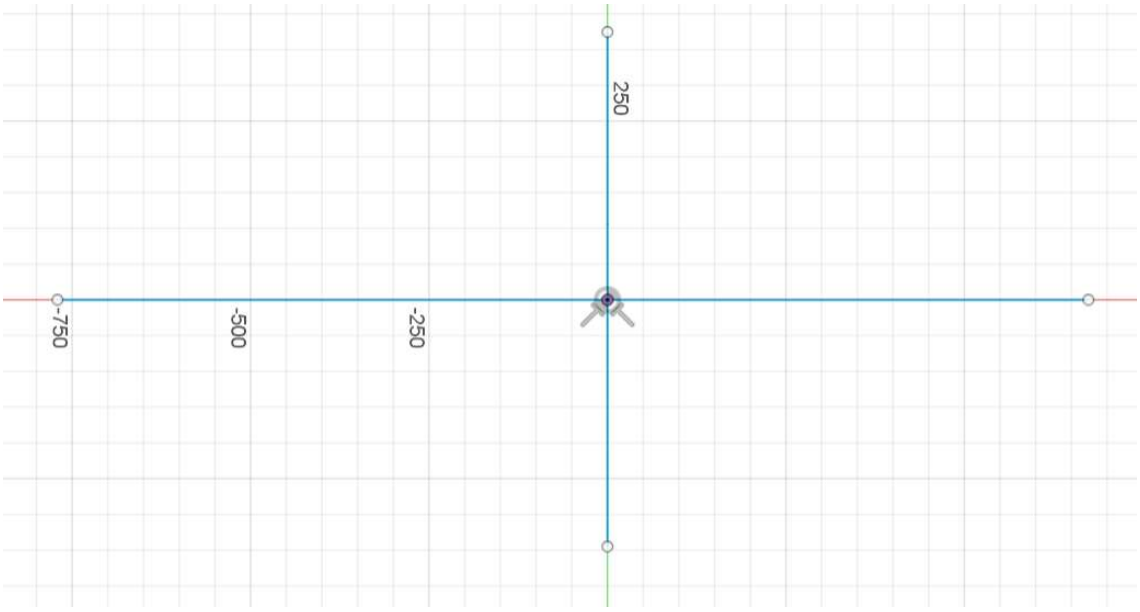
1.Initialize Workspace & Units:

Open Autodesk Fusion 360, create a new design, expand the **Document Settings** browser folder, and ensure the active units are set to **Millimeters (mm)**.

2.Create Centerlines & Construction Geometry:

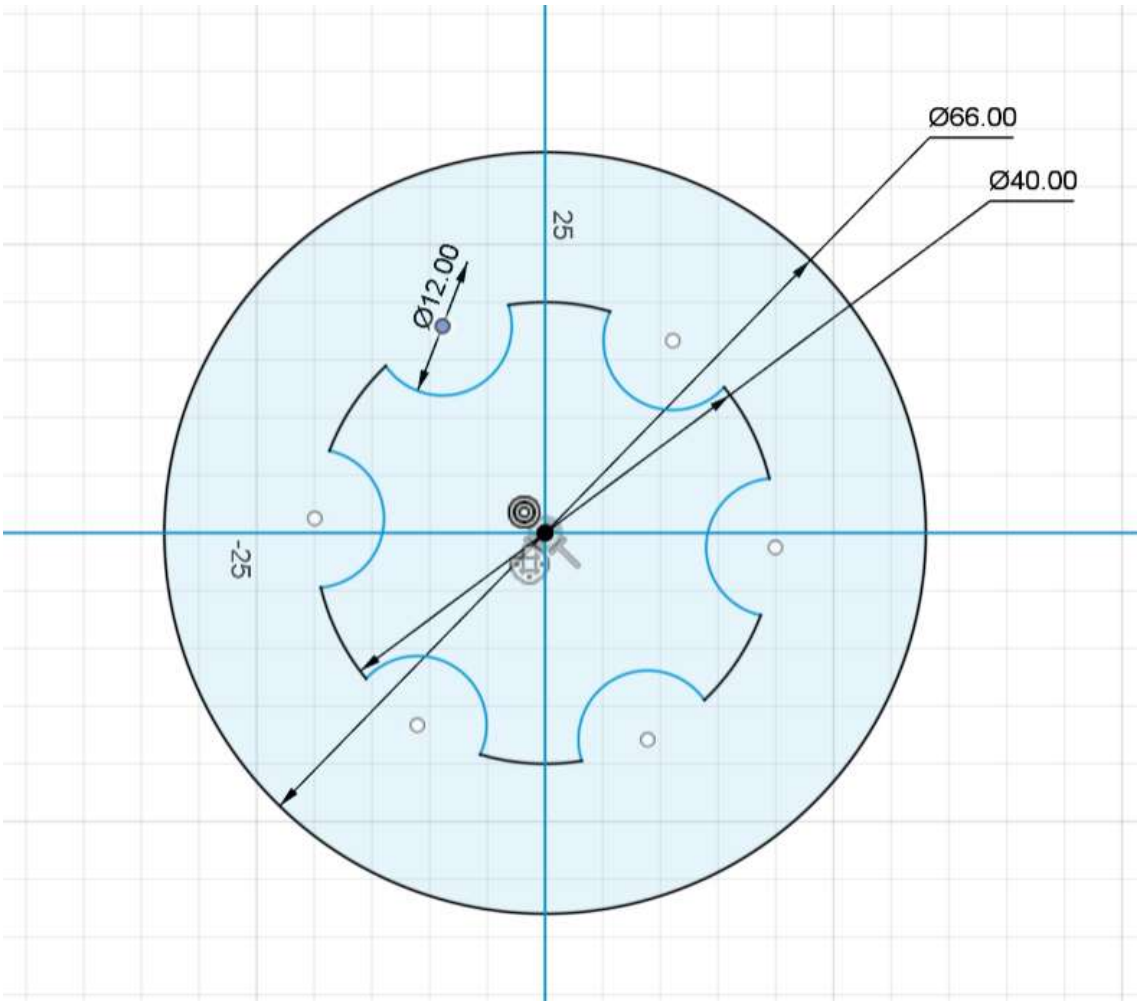
Click **Create Sketch** and select the XY plane. Use the Line tool combined with the **Construction** toggle to draw permanent vertical and intersecting horizontal reference

centerlines passing through the origin $(0, 0)$.



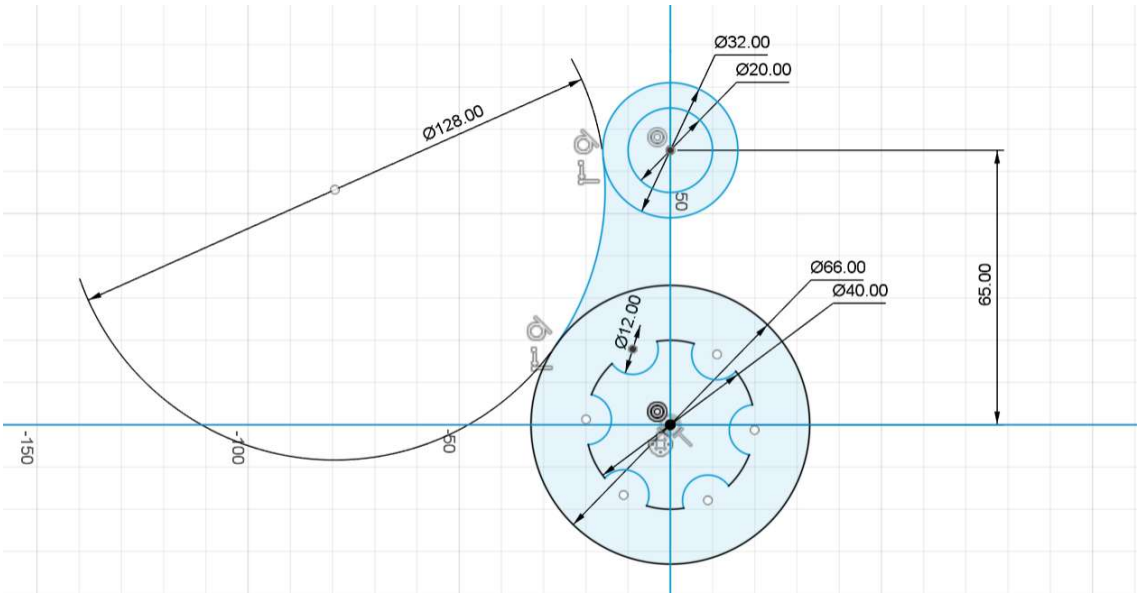
3.Model the Central Body Profile:

Using the Center Circle tool, draw a R20 circle at the origin. Create the 6 scalloped pockets around it using R6 arcs, trimming any intersecting inner profiles. Draw the outer R33 reference body arcs.



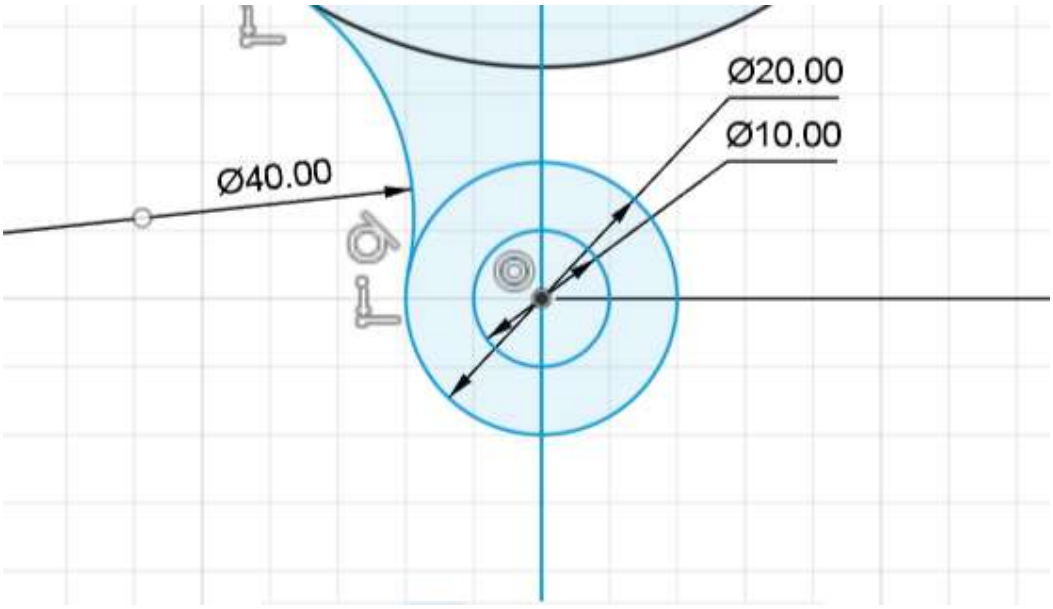
4.Construct the Top Lobe:

Draw a construction point or line straight up along the vertical axis at **65 mm** from the origin. At this center point, draw concentric circles of R10 and R16. Apply smooth **Tangent** constraints or R64 transition arcs connecting it to the central body.



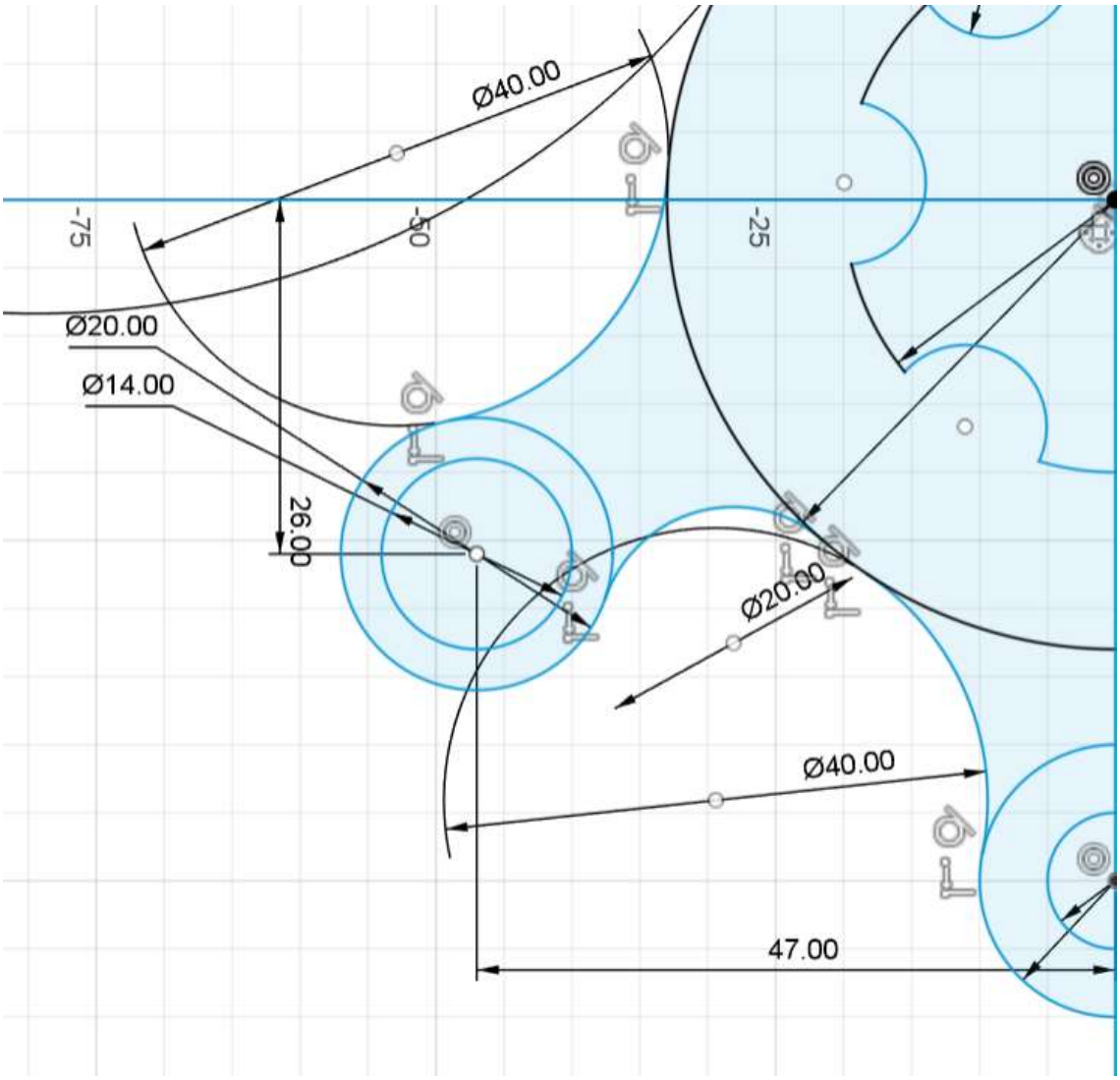
5.Construct the Bottom Lobe:

Measure down the vertical axis by a total distance of **50 mm** $(26mm + 24mm)$ to locate the bottom lobe center. Draw concentric circles for the R5 inner hole and R10 outer profile, joining them with R20 fillets.



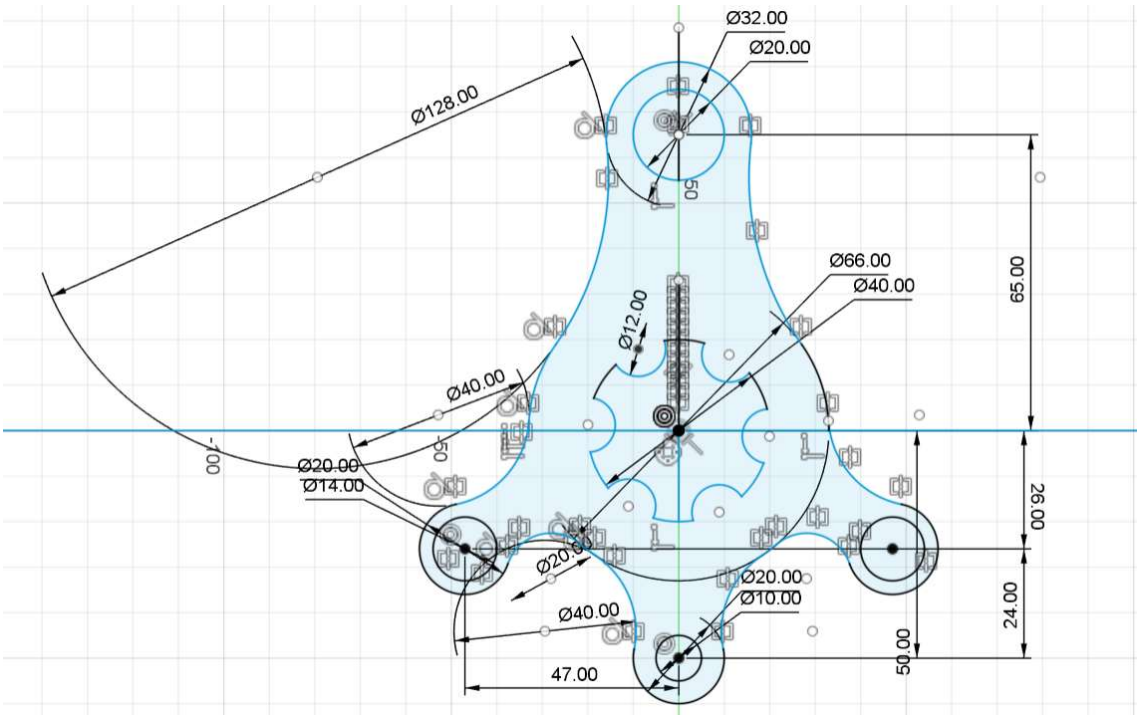
6.Construct Left and Right Side Lobes:

Draw construction lines offset **47 mm** horizontally and **26 mm** vertically downward from the center. At these coordinates, draw the side lobes containing an inner radius R7 and outer radius R10, then blend them into the main body using R10 and R20 transition arcs.



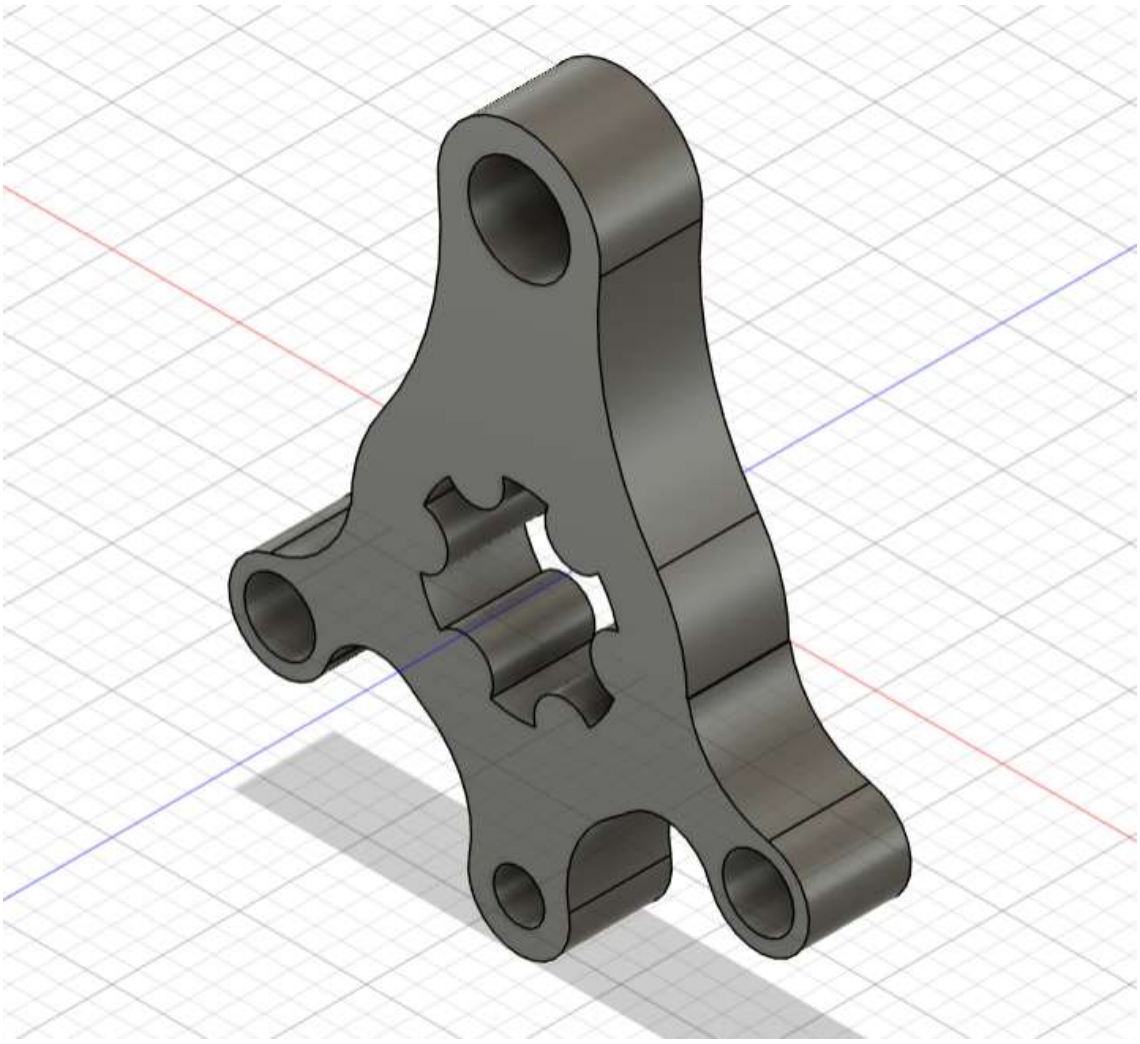
7.Trim and Fully Constrain the Sketch:

Use the **Trim** tool (Shortcut: T) to clear overlapping segments. Apply **Sketch Dimensions** (D) to lock down all radii, diameters, and offsets until all sketch lines turn black (indicating a fully constrained sketch).

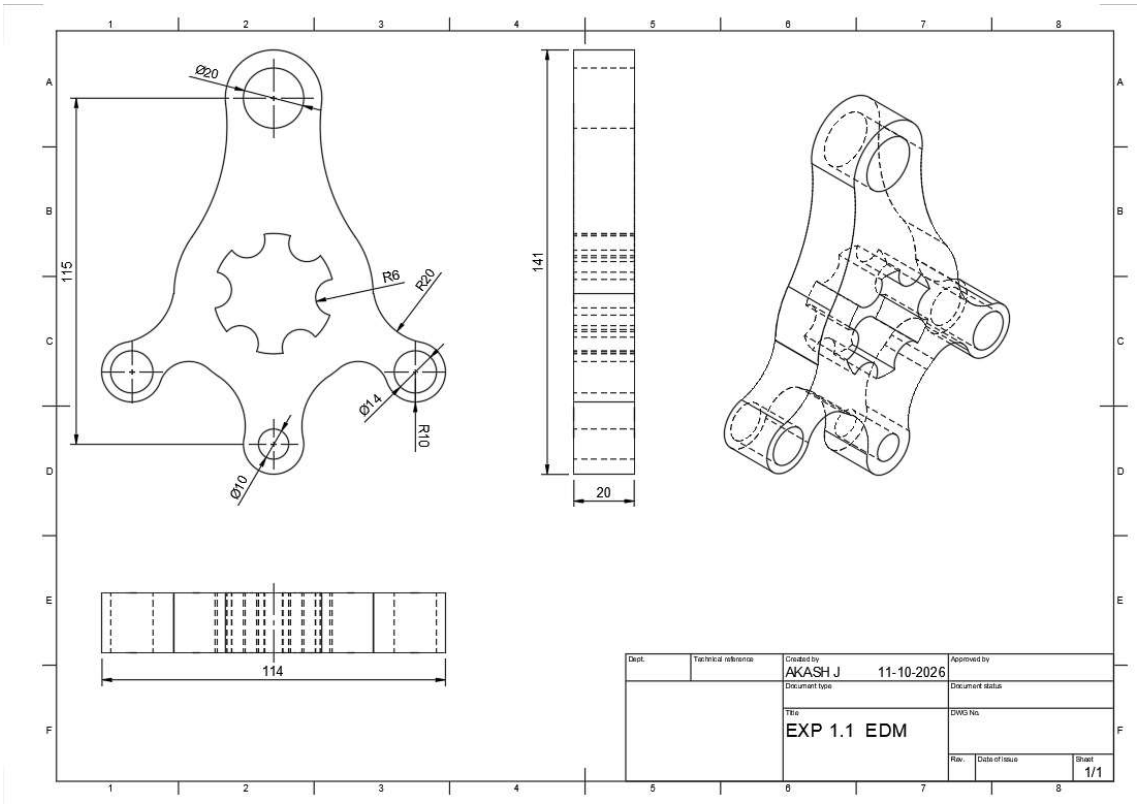


8.Finish Sketch and Extrude:

Click **Finish Sketch**. Select the **Extrude** tool, choose your closed profile, and input your desired component thickness (e.g., 5 mm or 10 mm) to generate the 3D solid model.



OUTPUT



RESULT

Thus the given sketch is drawn and drafted using fusion 360 tool.